

# Humayra Akter

Bangladesh



Email



GitHub



LinkedIn



Portfolio



Scholar

## Education

### BSc in Computer Science & Engineering

June 2025

Military Institute of Science and Technology (MIST), Bangladesh [<https://mist.ac.bd/>]

CGPA: 3.33 out of 4 [CGPA in the last 60 credits: 3.79 out of 4]

IELTS Academic: Overall 7.0 (L7, R7, W6.5, S6.5)

## Research Interests

Machine learning for real-world behavioral and health-related data, including multimodal sensing and speech/audio analytics, with an emphasis on interpretable, robust, and leak-aware modeling. I am particularly interested in human-centered AI and interactive systems that bridge machine learning research with deployable software solutions for practical impact.

## Undergrade Thesis

### Smart Pricing in Online Marketplaces: A Machine Learning and Analytics Framework [[DOI](#)]

**Summary:** Developed a machine learning based pricing framework for online marketplaces using extensive feature engineering and ensemble learning. Evaluated 14 regression models and proposed a stacked ensemble achieving  $R^2 = 0.8633$ ,  $RMSE = 303.83$ , and  $MAE = 215.88$  on laptop price prediction. The final model was deployed as a real-time Streamlit web application for practical decision support. *This work was later published at the STI-2025 conference.*

## Journal Articles

### ▪ Leakage-Audited Machine Learning for Childhood ADHD: Separating Symptom Overlap, Contextual Risk, and Diagnosis Discordance in National Survey Data

Manuscript under review– [Springer Nature]

**Summary:** Developed a leakage-audited machine learning framework to predict caregiver-reported current ADHD diagnosis using national child health survey data, separating diagnosis, treatment, symptom-overlap, demographic, and contextual variables before modeling. Instead of maximizing accuracy through possible information leakage, it shows how machine learning can support responsible population-level ADHD risk stratification and diagnosis-discordance analysis.

## Conference Papers

### ▪ LaundryMate: A React Native-Based Mobile Application for Smart Laundry Management [[DOI](#)]

Accepted at QPAIN-2025

### ▪ A Reproducible Baseline Study of Closed-Set Speaker Identification Using MFCCs and a Lightweight DNN on TIMIT [[DOI](#)]

Manuscript accepted at QPAIN-2026

### ▪ An IoT-Based Multisensor System for Real-Time Sleep Stage Monitoring [[DOI](#)]

Manuscript accepted at QPAIN-2026

### ▪ A Terrain-Aware Instance Segmentation Framework for Building-Level Flood Risk Mapping from High-Resolution Satellite Imagery

Manuscript under Preparation

## Technical Skills

**Programming Languages:** Python, C, C++, JavaScript, TypeScript

**Machine Learning & Data Analysis:** Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, Keras, OpenCV, MediaPipe (pose estimation), Matplotlib, Seaborn, signal preprocessing, feature extraction (MFCCs), statistical analysis, Jupyter Notebook, Google Colab

**Software & Systems Development:** React.js, Next.js, React Native, Node.js, Express.js, REST APIs, JWT, MongoDB, Oracle, Firebase, Streamlit, Power BI, Prisma, PostgreSQL

**DevOps & Tooling:** GitHub, Netlify, Vercel, Postman, Figma, LaTeX

## Experience

- **Analyst || TRPGLOBAL ([www.techriskpartners.com](http://www.techriskpartners.com))** 2026 - Present
  - Developed 4 full-stack web applications using modern web technologies within less than 3.5 months.
  - Designed and implemented a public-facing government law and risk-assessment platform.
  - Engineered scalable client-facing digital products, including one platform currently supporting multiple major enterprise clients.
  - Awarded Employee of the Month during the first month of permanent employment for rapid delivery, ownership, and engineering excellence.
- **Founder of MIST Career Club ([career-club.mist.ac.bd](http://career-club.mist.ac.bd))** 2024-2025
- **Frontend Volunteer, Web & IT Committee || IEEE Computer Society Bangladesh Chapter ([Link](#))** 2025
- **Instructor, Beginner-Level Web Development Course || MIST Career Club ([Link](#))** 2024

## Internship

**Bangladesh Computer Council (BCC)** 2024

Worked on the Election Commission's national ICT website: wrote functional test cases, validated UI/UX, and organized defects in test sheets. Exposure to national-scale systems, including switch/router configuration scripts, VLAN setups, routing logic, and government data storage infrastructure.

## Engineering Projects

- **Elite Dwell Assist ([Live Link](#) || [Project Demo](#))** 2023

*The project is about booking different home services for a time period or instance. Developed and deployed this full-stack web application using ReactJS, Node.js, MongoDB, and Firebase, focusing on reliability, authentication, and user-centric design.*
- **WiFi-based Human Activity Recognition (HAR) ( [Git Link](#) )** 2025

*Developed a HAR system using smartphone accelerometer and gyroscope data with engineered time-series features (561 to 150 via variance and correlation pruning). Trained ensemble tree models achieving 92.9% accuracy on unseen subjects, with strong performance on LAYING and minor confusion between SITTING and STANDING.*

## Achievements

- **Problem setter for National and Intra-Department Hackathons** 2025
- **1st Runner-up** in National Project Showcasing *MIST Intervention 4.1* 2025
- **1st Runner-up** in National Project Showcasing *BCS ICT Fest 2025* at BUET 2025
- **Finalist** at *WebXtreme* NSU inter-university software development *Hackathon* 2025
- **Finalist** at National Business Case Competition, *Nexus of Wit* at Rabindra University 2025
- **2nd Runner-up**, BUET CaseCraft AutoRealm national business case challenge 2024
- **Finalist** at the national robotics and tech competition, *Technoxian Bangladesh* 2024
- **Finalist** at Inter-University Software Competition, *DevSprint BUET* 2024
- **Top 23**, *Independence Day Programming Contest (IDPC)*, MIST 2023

## Extra Curricular

- **Club Roles || MIST Career Club** 2022-2025

*Led design and logistics for national-level career events and workshops.*
- **Vice President & R&D Committee Executive || MIST Computer Club** 2023-2025

*Led the official website development team, organized a hackathon, and mentored students on project building.*
- **Organizer || Shestem presents Engibiz National Event, MIST** 2024-2025

*Designed and led the creative design team and got the best sub-team prize of the event.*
- **Organizer || Shestem presents MIST Job Fair, 2025** 2025

*Designed and led the design team for the event.*

## Reference

### T. M. Shahriar Sazzad

Associate Professor (Project Supervisor)  
Military Institute of Science and Technology  
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### M. Akhtaruzzaman

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